

Richard Medina Rodríguez

Curriculum Vitae – May 2026

CEREMADE, Université Paris Dauphine - PSL
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Research Profile

Analysis of PDEs in kinetic theory and statistical physics, with emphasis on boundary effects, domain geometry, and long-time behaviour of solutions.

Current direction: modeling and numerical simulation of complex systems, using my analytical background to guide model construction and support the justification of numerical approaches.

Publications

Accepted/Published

- 2026 **Evans, J., Medina, R.**, *Existence and Stability of Non-Equilibrium Steady States of a Weakly Non-Linear Kinetic Fokker-Planck Equation in a Domain*, J Stat Phys 193, 47 [doi](#).
- Existence and asymptotic stability of stationary solutions to a nonlinear kinetic Fokker-Planck equation with non-equilibrium forcing via BGK thermostats and non-isothermal Maxwell boundary conditions.

Preprints

- 2026 **R. Medina**, The Boltzmann equation with non-isothermal Maxwell boundary conditions, URL <https://arxiv.org/abs/2604.13572>.
- Existence, uniqueness and asymptotic stability of non-equilibrium steady states for the Boltzmann equation in non-isothermal bounded domains, under small boundary temperature fluctuations.
- 2025 **R. Medina**, The Boltzmann equation on smooth and cylindrical domains with Maxwell boundary conditions, URL <https://arxiv.org/abs/2510.13260>. Submitted to *Kinetic and Related Models*.
- Well-posedness and exponential convergence of solutions for a Boltzmann equation in bounded domains, including smooth and cylindrical geometries, with Maxwell boundary conditions.
- 2024 **K. Carrapatoso, P. Gabriel, R. Medina, and S. Mischler**, Constructive Krein-Rutman result for Kinetic Fokker-Planck equations in a domain, URL <https://arxiv.org/abs/2407.10530>. Submitted to *Analysis & PDE*.
- Well-posedness and exponential convergence to equilibrium for general kinetic Fokker-Planck equations in non-isothermal bounded domains.

Long-term Research Stays

- Oct. – Nov. 2024 **Warwick Mathematics Institute**, University of Warwick, United Kingdom.
Work in collaboration with Josephine Evans on the effects of interior and boundary thermostats on the long-time behavior of non-linear kinetic Fokker-Planck equations.

Education and Academic Positions

- 2025 – 2026 **Teaching and Research Fellow (A.T.E.R.)**, Université Paris Dauphine - PSL, France.
Research on kinetic theory; teaching assistant for mathematics courses (real analysis and numerical methods).

- 2022 – 2025 **PhD in Applied Mathematics**, Université Paris Dauphine - PSL, France.
 Thesis title: Hypocoercivity and geometric confinement.
 PhD Advisors: Kleber Carrapatoso & Stéphane Mischler.
 Defense date: 18 December 2025.
 Thesis reviewers: Véronique BAGLAND & José A. CAÑIZO.
 Thesis examiners: Émeric BOUIN, Arnaud GUILLIN & Frédéric HÉRAU (president of the jury).
- 2022 **Master Internship**, CEREMADE, Université Paris Dauphine - PSL, France.
 Advisors: Kleber Carrapatoso & Stéphane Mischler.
 Research Project: Extending hypocoercivity techniques for linearized kinetic equations to non-smooth bounded domains.
- 2021 – 2022 **Master of Science in Applied and Theoretical Mathematics (M2 MATH)**, Université Paris Dauphine - PSL, France.
 With Honors (Mention Bien).
- 2020 – 2021 **Assistant Professor**, *Department of Mathematics*, Universidad de La Habana, Cuba.
 Research on functional analysis and PDEs; teaching assistant for mathematics courses (real and complex analysis).
- 2016 – 2020 **Bachelor of Science in Mathematics**, Universidad de La Habana, Cuba.
 Summa Cum Laude.
 Bachelor's Final Project: Random Schrödinger operators in a 2D scheme.
 Advisors: Rita Roldán Inguanzo & Laure Dumaz.

Selected Invited Talks

Invited Talks at Conferences and Workshops

- Jun. 2024 ***The Boltzmann equation in a cylinder near the hydrodynamic limit***, presented at the conference *Analysis of PDEs in Mathematical Physics*, University of Bath, United Kingdom.
- Feb. 2023 ***Weighted L^∞ estimates for the Boltzmann equation***, presented at the *MAFRAN Days*, King's College, United Kingdom.
- Jun. 2019 ***Fixed points in non-expansive maps***, presented at the *International Congress COMPUTAT 2019*, Universidad de La Habana, Cuba.
- Mar. 2019 ***Extrapolation Methods for Approximating a Fairness Functional***, presented at the *International Workshop on Operations Research (IWOR)*, Universidad de La Habana, Cuba.
- Mar. 2018 ***Restriction-based interpolation with cubic A-splines***, presented at the *International Conference on Operations Research (ICOR)*, Universidad de La Habana, Cuba.

Invited Talks at Seminars

- Feb. 2026 ***The Boltzmann equation with non-isothermal Maxwell boundary conditions***, presented at the *SPIKE Seminar*, Institut Henri Poincaré (IHP), France.
- Jan. 2025 ***Brief introduction to the Boltzmann equation***, presented at the *Young Researchers Seminar*, Université Paris Dauphine - PSL, France.
- Nov. 2024 ***Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler)***, presented at the *Junior Analysis Seminar*, Imperial College London, United Kingdom.
- Oct. 2024 ***The Boltzmann equation on C^1 and cylindrical domains near the hydrodynamic limit***, presented at the *Partial Differential Equations and their Applications Seminar*, University of Warwick, United Kingdom.
- Nov. 2023 ***Introduction to the kinetic Fokker-Planck equation and its long-time behavior***, presented at the *Young Researchers Seminar*, Université Paris Dauphine - PSL, France.
- May 2023 ***Hypocoercivity estimates for some linearized kinetic operators***, presented at the *Young Researchers Seminar*, Université de Lille, France.

- Mar. 2023 ***Hypocoercivity estimates for some linearized kinetic operators***, presented at the *Young Researchers Seminar*, Université Paris Dauphine - PSL, France.

Poster Presentations

- Jul. 2024 ***Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler)***, presented at the summer school *Collective behavior and Pattern formation*, CIRM, France.
- Jun. 2024 ***Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler)***, presented at the summer school *Frontiers in Interacting Particle Systems, Aggregation-Diffusion Equations & Collective Behavior*, CIRM, France.

Participation on Conferences, Workshops and Schools

- Jul. 2024 ***Summer school on Collective Behavior and Pattern Formation***, CIRM, France.
- Jun. 2024 ***Research school on Frontiers in Interacting Particle Systems Aggregation-Diffusion Equations & Collective Behavior***, CIRM, France.
- Nov. 2022 ***Research School on Kinetic Theory***, CIRM, France.
- Jul. 2022 ***Conference on When Kinetic Theory Meets Fluid Mechanics***, FIM - Institute for Mathematical Research, ETH Zürich, Switzerland.
- Jun. 2019 ***CIMPA Research Summer School on Mathematical Models in Biology and Related Applications of Partial Differential Equations***, Universidad de La Habana, Cuba.
- Mar. 2019 ***III Winter School in Computer Science on Selected Topics in Probability and Statistics***, Universidad de La Habana, Cuba.

Awards, Fellowships & Grants

- 2022 ***MathInParis2020 COFUND Fellowship*** — PhD grant awarded by the Fondation de Sciences Mathématiques de Paris (FSMP).
- 2021 ***PSL PhD Track Grant 2021*** — Master's Excellence Scholarship awarded by the Université Paris Sciences et Lettres.
- 2020 ***Prize for Scientific Merit*** awarded by the Universidad de La Habana for outstanding scientific work during the Bachelor's studies.
- 2018 Recipient of ***SciGenome Research Foundation (SGRF) GYAN Scholarship*** to participate ***Nextgen Genomics, Biology, Bioinformatics and Technologies-2018*** meeting at Jaipur India from 30th September to 2nd October 2018.

Organization of Seminars and Conferences

- Jun. 2024 Co-organizer of the ***Young Researchers Days*** conference, held at *Domaine de la Tour*, Saint-Pierre-Canivet, France. Three-day conference for PhD students to present and discuss their research. Supported by the Dauphine Doctoral School.
- 2023 – 2024 Co-organizer of the ***Young Researchers Seminar*** at the Université Paris Dauphine - PSL.

Teaching Experience

- Spring 2026 ***Algebra 4 & Numerical methods***, problem-solving sessions for second-year Mathematics students, Université Paris Dauphine - PSL, France.
- Fall 2025 ***Analysis 1***, lectures and problem-solving sessions for first-year Mathematics students, Université Paris Dauphine - PSL, France.

- May 2025 **Functional Analysis Summer School**, *problem-solving sessions for graduate Mathematics students*, Universidad de La Habana, Cuba.
- Spring 2024 & 2025 **Analysis 2**, *problem-solving sessions for first-year Mathematics students*, Université Paris Dauphine - PSL, France.
- Fall 2023 **Mathematical Methods**, *lectures and problem-solving sessions for first-year Economics students*, Université Paris Dauphine - PSL, France.
- Spring 2021 **Complex Analysis**, *problem-solving sessions for third-year Mathematics students*, Universidad de La Habana, Cuba.
- Fall 2020 **Introduction to Mathematics**, *lectures and problem-solving sessions for first-year Mathematics students*, Universidad de La Habana, Cuba.
- 2017 – 2019 **Tutorship**, *assisting in problem-solving sessions for Mathematics students*, Universidad de La Habana, Cuba.
Subjects: Topology, Analysis 4 (series and Fourier analysis), Analysis 3 (analysis of functions of several variables), Analysis 2 (analysis of functions of one variable), and Introduction to Mathematics.

Outreach

- May 2024 Volunteer Stand Animator at the **Salon de Culture et Jeux Mathématiques** at Place Saint-Sulpice, Paris, France.
- March 2024 Volunteer judge at the **Math en Jeans 2024 Congress** at the Université Paris Dauphine - PSL, France.
- 2016 – 2019 Instructor for high-school students in an advanced mathematics program, preparing them for the Cuban National Mathematics Competitions, La Habana, Cuba.

Editorial Service

Referee for international journals (*Communications in Mathematical Physics*).

Computer Skills

- Programming Languages PYTHON (advanced), C# (intermediate), HTML (intermediate).
- Scientific software WOLFRAM MATHEMATICA (advanced), MATLAB (intermediate).
- Typesetting & visualization L^AT_EX (advanced), TikZ/Beamer (advanced).

Competitions

- 2019 **Ibero-American University Mathematics Olympiad** - Silver medal.
- 2018 **Ibero-American University Mathematics Olympiad** - Bronze medal.
- 2017 **Raimundo Reguera National University Mathematics Olympiad** - Silver medal.
- 2017 **Ibero-American University Mathematics Olympiad** - Honorable Mention.
- 2016 **Raimundo Reguera National University Mathematics Olympiad** - Honorable Mention.
- 2015 **Cuban National High School Mathematics Olympiad** - Gold medal (ranked 3rd).
- 2014 **Cuban National High School Mathematics Olympiad** - Silver medal.

Languages

- Spanish Native speaker.
- English Bilingual proficiency, TOEIC C1 (June 2022).
- French Bilingual proficiency.

Referees

Prof. Stéphane Mischler

*CEREMADE, Université Paris Dauphine - PSL;
& Institut Universitaire de France (IUF).*

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Prof. Kleber Carrapatoso

*Centre de Mathématiques Laurent Schwartz (CMLS),
École Polytechnique.*

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Prof. Josephine Evans

*Warwick Mathematics Institute,
University of Warwick.*

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Prof. Pierre Gabriel

*Institut Denis Poisson,
Université de Tours.*

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